

TOPS TECH Task 8 (CS)

Task-9 Vulnerability Scanning Submitted

- 1. Install and run the free version of the OpenVAS or Nessus vulnerability scanner on your system, and perform a basic scan of your own computer or a local virtual machine.**

Ans:

Vulnerability Scanning Using Nessus Essentials

Objective:

- To identify security vulnerabilities, misconfigurations, and potential risks present on the local system using Nessus Essentials Vulnerability Scanner.
- Open Nessus through the web interface using:
<https://localhost:8834>
- Create a new scan by selecting "Basic Network Scan".

Enter the scan details:

- Name: Localhost Scan
- Target: 127.0.0.1
- Save the scan configuration and launch the scan.
- Wait for the scan to complete and review the generated report.

Observations:

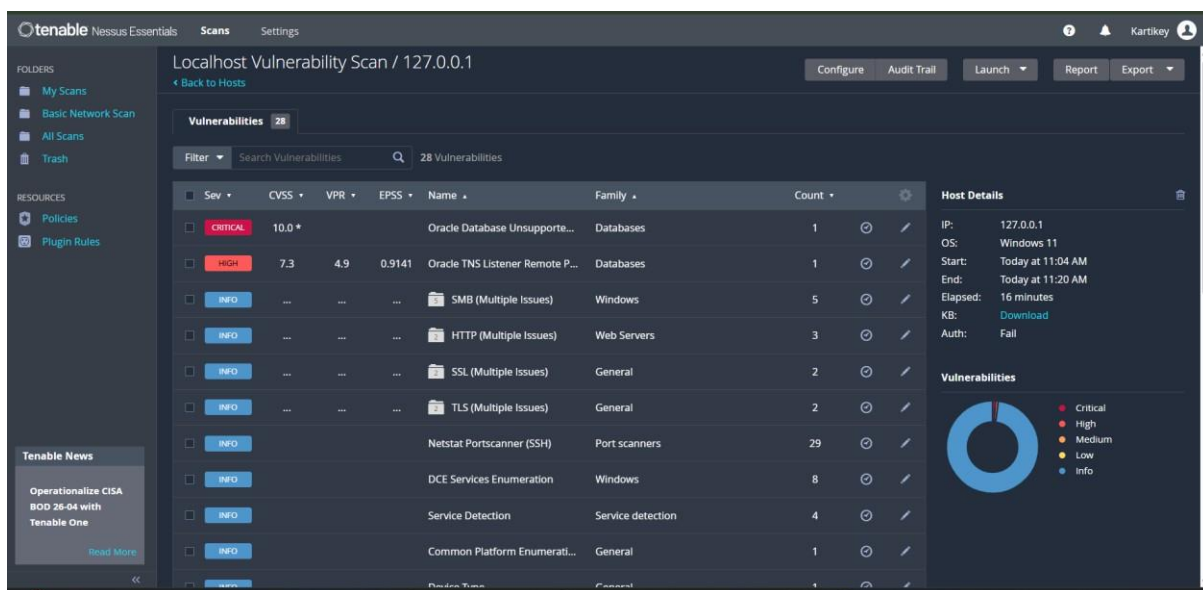
- Nessus successfully scanned the local system and identified active services and system configurations.
- The scanner checked installed software, operating system configuration, open ports, and security settings.
- Several informational items and warnings were detected during the scan.
- Nessus generated a detailed report showing vulnerabilities categorized by severity levels such as Critical, High, Medium, Low, and Informational.

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- The scan report also provided recommendations for improving system security.

Sample Vulnerabilities Identified:

- Severity Description
 - Medium Outdated software version detected
 - Low SSL/TLS configuration warning
 - Info Open network service identified
- ⇒ Nessus Essentials was successfully installed and configured.
- ⇒ A vulnerability assessment of the local system was performed successfully.
- ⇒ The scan report identified system information, open services, and potential security risks that can be used to improve the overall security posture of the system.



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2. Scan your Wi-Fi router's IP address using Nmap and list all open ports detected.

Hint: Use the command 'nmap -v [router_ip]' in your terminal or command prompt.

Ans:

Wi-Fi Router Port Scanning Using Nmap

Objective:

- To identify open ports and associated network services running on the Wi-Fi router using the Nmap network scanning tool.

Command:

- ipconfig
- The ipconfig command is used to identify the Default Gateway address, which represents the IP address of the Wi-Fi router.

Example:

- Default Gateway : 192.168.31.1

Command:

- nmap -sV 192.168.31.1
- The nmap -sV command scans the target device and attempts to determine the services and versions running on open ports.

Observations:

- Nmap successfully scanned the Wi-Fi router and identified several open ports.
- The detected ports were associated with router management and network communication services.
- Open ports indicate services currently accepting network connections.

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The following open ports were identified:

<u>Port Number</u>	<u>Protocol</u>	<u>State</u>	<u>Service</u>
53	TCP	OPEN	domain
80	TCP	OPEN	http
443	TCP	OPEN	https
8080	TCP	OPEN	http-proxy
8443	TCP	OPEN	https-alt

Port Description:

<u>Port Number</u>	<u>Description</u>
53	Domain Name Service (DNS)
80	Web-based Router Management Interface
443	Secure Router Management Interface
8080	Alternative HTTP Management Service
8443	Alternative Secure HTTPS Management Service

- The router was found to be running network management services required for administration and internet connectivity.
 - Ports 8080 and 8443 indicate additional web-based management interfaces available on the router.
 - No critical issues were observed during the basic port scan.
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- ⇒ The Wi-Fi router was successfully scanned using Nmap.
 - ⇒ Open ports and associated services were identified and documented.
 - ⇒ The scan provided useful information regarding the network services exposed by the router and helped understand the network attack surface.

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3. **Analyze the scan report from your vulnerability scanner and identify at least 3 vulnerabilities or warnings it reports; for each, write a one-line description of what the risk is.**

Ans:

Vulnerability Analysis Using Nessus Scan Report

Objective:

- To analyze the Nessus vulnerability scan report and identify vulnerabilities and security warnings present on the local system.

Tool Used:

- Nessus Essentials Vulnerability Scanner

Target:

- 127.0.0.1 (Localhost)

Observations:

- Nessus successfully scanned the local system and identified 28 vulnerabilities and informational findings.
- The scan report categorized findings into Critical, High, and Informational severity levels.
- Several issues related to Oracle services, SMB services, HTTP services, SSL/TLS configurations, and service enumeration were detected.

The following vulnerabilities/warnings were identified:

1. Oracle Database Unsupported Version

- Severity: Critical
- Risk:
 - ➔ The Oracle Database version installed on the system is no longer supported by the vendor.
 - ➔ Unsupported software does not receive security updates and may contain known vulnerabilities that attackers can exploit.

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Recommendation:

- Upgrade to a supported Oracle Database version and apply the latest security patches.

2. Oracle TNS Listener Remote Information Disclosure

- Severity: High
- Risk:
 - ➔ The Oracle TNS Listener may disclose sensitive information about the database environment, which can assist attackers in gathering intelligence for further attacks.

Recommendation:

- Restrict listener access and apply Oracle security updates.

3. SMB (Multiple Issues)

- Severity: Informational
- Risk:
 - ➔ SMB services may expose information about the system configuration and network resources, increasing the attack surface if improperly

Recommendation:

- Disable unnecessary SMB services and restrict access using firewall rules.

4. HTTP (Multiple Issues)

- Severity: Informational
- Risk:
 - ➔ Web services may reveal server information, configuration details, or application fingerprints that can aid attackers during reconnaissance.

Recommendation:

- Minimize information disclosure and keep web services updated.

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4. SSL/TLS (Multiple Issues)

- Severity: Informational
- Risk:
 - ➔ Weak SSL/TLS configurations may expose encrypted communications to interception or downgrade attacks.

Recommendation:

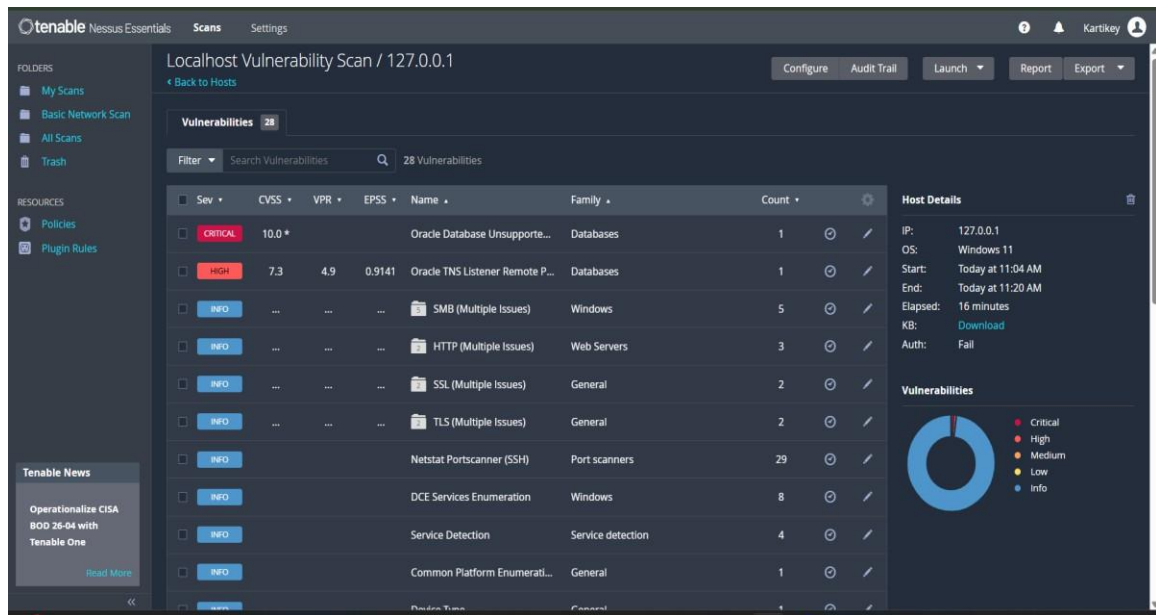
- Use strong encryption protocols and disable outdated SSL/TLS versions.

Summary Table:

Vulnerability Severity Risk Description

- Oracle Database Unsupported Version Critical Unsupported software may contain exploitable vulnerabilities.
 - Oracle TNS Listener Information Disclosure High Sensitive system information may be exposed.
 - SMB Multiple Issues Informational May reveal system and network details.
 - HTTP Multiple Issues Informational May expose web server information.
 - SSL/TLS Multiple Issues Informational Weak encryption settings may affect communication security.
-
- ⇒ The Nessus scan report was successfully analyzed.
 - ⇒ One Critical vulnerability, one High severity vulnerability, and multiple informational security warnings were identified.
 - ⇒ Appropriate recommendations were suggested to reduce security risks and improve the overall security posture of the system.

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4. Pick any app you use daily (like Instagram, Zomato, or Paytm) and research one real vulnerability that was found in it in the past; write a short summary of what the vulnerability was and how it could have been detected by scanning tools.

Ans:

Research on a Real Vulnerability in WhatsApp

Objective:

- To study a real-world security vulnerability discovered in a widely used application and understand its impact and detection methods.

Application Selected:

- WhatsApp

Vulnerability Name:

- Pegasus Spyware Vulnerability (2019)

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- In 2019, a serious vulnerability was discovered in WhatsApp that allowed attackers to install Pegasus spyware on a victim's mobile device.
- The vulnerability existed in WhatsApp's voice calling feature.
- An attacker could exploit this flaw by placing a specially crafted WhatsApp call to the target device.
- In some cases, the spyware could be installed even if the victim did not answer the call.

Impact:

- Attackers could gain unauthorized access to sensitive information stored on the device.
- Personal messages, emails, contacts, location data, and other private information could potentially be monitored.
- The vulnerability affected both Android and iOS devices.

Risk Analysis:

Risk Type Description:

- Privacy Risk User data could be accessed without permission.
- Information Disclosure Sensitive information could be exposed.
- Remote Exploitation Attackers could compromise a device remotely.
- Surveillance Risk Spyware could monitor user activities secretly.

How Scanning Tools Could Detect It:

- Vulnerability scanners can identify outdated application versions that contain known security flaws.
- Security assessment tools can compare installed software versions against vulnerability databases.
- Mobile security scanners can detect suspicious application behaviour and unusual network activity.
- Regular security audits and vulnerability assessments help identify software that requires security updates.

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Control:

- Update WhatsApp to the latest version.
 - Install security patches released by the vendor.
 - Avoid using outdated software versions.
 - Enable device security features and keep the operating system updated.
- ⇒ A real-world vulnerability affecting WhatsApp was successfully studied.
- ⇒ The Pegasus spyware vulnerability demonstrated how a software flaw can lead to serious privacy and security risks.
- ⇒ Regular updates, vulnerability scanning, and security monitoring can help prevent such attacks and improve overall cybersecurity.

- 5. Use ChatGPT or Copilot to generate a basic bash or PowerShell script that runs Nmap on a list of 3 IP addresses and saves the results to separate files; copy and run the script, and share the output files.

Hint: Ask the AI tool for a script that loops through IPs and uses Nmap for each.**

Ans:

Automated Nmap Scanning Using PowerShell Script

Objective:

- To automate Nmap scanning of multiple IP addresses using a PowerShell script and store the results in separate output files.

Tool Used:

- Nmap
- Windows PowerShell

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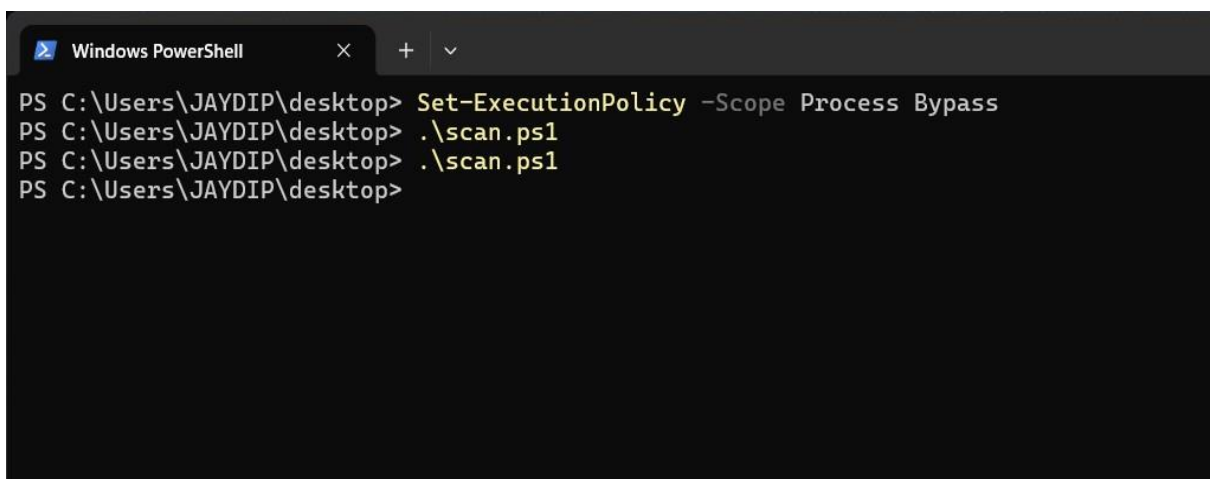
PowerShell Script:

```
$ips = @(
"192.168.31.1",
"127.0.0.1",
"192.168.31.165"
)

foreach ($ip in $ips)
{
nmap -F $ip > "$ip.txt"
}
```

Script Explanation:

- The script stores three IP addresses in an array.
- The foreach loop processes each IP address one by one.
- Nmap performs a fast scan (-F) on each target.
- The output of each scan is redirected to a separate text file.
- The generated files are named according to the scanned IP address.



```
Windows PowerShell
PS C:\Users\JAYDIP\desktop> Set-ExecutionPolicy -Scope Process Bypass
PS C:\Users\JAYDIP\desktop> .\scan.ps1
PS C:\Users\JAYDIP\desktop> .\scan.ps1
PS C:\Users\JAYDIP\desktop>
```

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Observations:

- The script successfully automated the scanning process.
 - Each target was scanned individually.
 - Scan results were stored in separate files for easier analysis.
 - The generated reports contained information about open ports and available network services.
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- ⇒ A PowerShell script was successfully generated using ChatGPT.
 - ⇒ Nmap scans were executed automatically on multiple IP addresses.
 - ⇒ Individual scan reports were generated and stored in separate output files, demonstrating basic automation of network scanning tasks.